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Assignment Questions – Module 2

1. Brief about SplitMAC architecture and how it improves the AP's performance

Answer:

SplitMAC architecture divides the MAC (Medium Access Control) layer functions between the Access Point (AP) and the Wireless LAN Controller (WLC). In traditional autonomous APs, all MAC functions are handled locally at the AP. SplitMAC changes this by splitting the responsibilities:

- Real-time MAC functions (handled at AP): These include time-sensitive tasks such as beacon transmission, acknowledgment of frames, probe responses, power save management, and frame queuing. Since these require immediate response, they stay at the AP.
- Non-real-time MAC functions (handled at WLC): These include association, authentication, roaming decisions, security policy enforcement, and configuration management. These are centralized at the controller.

How it improves AP performance:

- Reduces the processing load on the AP — the AP only handles critical real-time tasks.
- The WLC handles complex decisions, allowing the AP to focus on wireless transmission.
- Simplifies AP hardware since it doesn't need to run full management logic.
- Enables centralized management, easier firmware updates, and consistent policy enforcement across all APs.
- Improves scalability — adding new APs requires minimal configuration since the WLC handles everything centrally.

2. Describe about CAPWAP, explain the flow between AP and Controller

Answer:

CAPWAP stands for Control And Provisioning of Wireless Access Points. It is a standard protocol defined in RFC 5415 that enables a WLC to manage a collection of wireless APs. It provides a standardized way for APs and controllers to communicate regardless of the vendor.

CAPWAP operates using two tunnels between the AP and the WLC:

- Control Tunnel (UDP Port 5246): Used for management traffic — configuration, firmware updates, AP state, statistics, and control messages. This tunnel is encrypted using DTLS (Datagram Transport Layer Security).
- Data Tunnel (UDP Port 5247): Used for forwarding client data frames from the AP to the WLC (in centralized mode). This tunnel is optionally encrypted.

Flow between AP and Controller:

1. **AP Discovery: When an AP powers on, it sends a CAPWAP Discovery Request to find available WLCs (via broadcast, multicast, or DNS).**
2. **WLC Response: The WLC replies with a Discovery Response containing its information.**
3. **DTLS Handshake: The AP initiates a DTLS session with the chosen WLC for secure communication.**
4. **Join Request/Response: The AP sends a Join Request; the WLC authenticates it and sends a Join Response.**

5. Configuration: The WLC pushes configuration (SSID, channels, policies) to the AP via the control tunnel.

6. Image Download: If firmware mismatch, the AP downloads the correct image from the WLC.

7. Run State: The AP enters Run state and starts serving clients. Client data is forwarded through the data tunnel to the WLC.

3. Where does CAPWAP fit in the OSI model, what are the two tunnels in CAPWAP and their purpose

Answer:

CAPWAP in the OSI Model:

CAPWAP operates at Layer 3 (Network Layer) of the OSI model. It uses UDP as the transport protocol (Layer 4) and runs over IP networks. This means APs and WLCs can be on different subnets or even geographically separated, as long as they have IP connectivity.

- Layer 3 (Network Layer): CAPWAP uses IP addressing for AP-WLC communication.
- Layer 4 (Transport Layer): Uses UDP — port 5246 for control and port 5247 for data.
- Layer 5–7: CAPWAP itself provides application-level control and management logic.

The Two Tunnels:

1. Control Tunnel (UDP 5246):

Purpose: Carries all management and control traffic between the AP and WLC. This includes AP configuration, firmware updates, statistics reporting, keep-alive messages, and state information. This tunnel is always DTLS encrypted to ensure secure management communication.

2. Data Tunnel (UDP 5247):

Purpose: Carries the actual client data frames. In centralized (local) mode, all wireless client traffic is encapsulated and forwarded to the WLC through this tunnel. The WLC then forwards it to the wired network. This tunnel is optionally encrypted depending on deployment needs.

4. What's the difference between Lightweight APs and Cloud-based APs

Answer:

Lightweight APs (LWAPP / CAPWAP-based):

- Managed by an on-premises Wireless LAN Controller (WLC) using CAPWAP protocol.
- The WLC handles configuration, security policies, roaming, and firmware updates.
- APs have minimal intelligence — they rely on the WLC for most decisions.
- Low latency since the controller is local to the network.
- Requires investment in WLC hardware.
- Suitable for enterprises with dedicated IT staff and on-site infrastructure.
- Example: Cisco Aironet APs with Cisco WLC.

Cloud-based APs:

- Managed via a cloud-hosted controller/dashboard accessible over the internet.
- APs are more intelligent — they can function independently even if cloud connectivity is lost (in most deployments).
- Configuration, monitoring, and updates are done remotely through a web portal.
- Easier to deploy across multiple sites with no need for on-premises controller hardware.
- Subscription-based model — ongoing cost but lower upfront investment.

• Suitable for distributed enterprises, retail chains, or businesses without dedicated IT teams.
• Example: Cisco Meraki, Aruba Central, Ubiquiti UniFi.

Key Difference Summary:

• Lightweight APs depend on a physical WLC on-premises; Cloud APs depend on an internet-connected cloud platform.

• Cloud APs are more flexible for multi-site deployments; Lightweight APs offer more control in single-site enterprise setups.

5. How is the CAPWAP tunnel maintained between AP and controller

Answer:

The CAPWAP tunnel between the AP and WLC is maintained through several mechanisms:

1. **DTLS Session:**
The control tunnel is secured using DTLS (Datagram Transport Layer Security). This session is established during the initial join process and kept alive as long as the AP is connected. If the DTLS session drops, the AP will re-initiate the discovery and join process.
2. **Keepalive Messages (Heartbeats):**
The AP and WLC exchange periodic Echo Request and Echo Response messages over the control tunnel. These act as heartbeats to confirm both ends are alive and reachable. If the AP misses a set number of consecutive keepalive responses, it considers the WLC unreachable.
3. **Retransmission and Timers:**
CAPWAP uses retransmission timers for control messages. If an acknowledgment is not received within the timeout period, the message is retransmitted up to a configured number of times.
4. **AP Failover:**
If the primary WLC becomes unreachable (keepalives fail), the AP will automatically try to connect to a secondary or tertiary WLC if configured. This ensures continuity of service.
5. **Data Tunnel Maintenance:**
The data tunnel (UDP 5247) runs in parallel and carries client traffic. Its continuity depends on the control tunnel remaining active. If the control tunnel fails, the data tunnel also goes down.
6. **Re-discovery on Failure:**
If all configured WLCs are unreachable, the AP enters a standalone or disconnected state. Depending on the AP mode (FlexConnect or Local), it may continue serving clients locally until the WLC is reachable again.

6. What's the difference between Sniffer and Monitor mode, use case for each mode

Answer:

Sniffer Mode:

• In Sniffer mode, the AP captures all wireless packets on a specific channel and forwards them to a packet analysis tool (such as Wireshark) running on a remote PC.

• The AP acts as a wireless packet capture device — it does not serve any clients.

• The captured frames include raw 802.11 headers and payload, making it useful for deep packet inspection.

• Use Cases:

- Troubleshooting client connectivity issues (authentication failures, association problems).

- Diagnosing protocol-level problems in the wireless network.
- Security audits to inspect unencrypted traffic.
- Analyzing specific channel behavior.

Monitor Mode:

• In Monitor mode, the AP passively scans all channels to detect rogue APs, rogue clients, and wireless intrusions.
• The AP does not serve any clients. It continuously scans and reports findings to the WLC.
• Used as part of a Wireless Intrusion Prevention System (WIPS).
• Use Cases:
- Detecting and locating rogue APs in the network.
- Identifying unauthorized clients trying to connect.
- Spectrum monitoring across all channels.
- IDS/IPS functionality for wireless networks.

Key Difference:

• Sniffer mode captures packets on one channel and sends them to an external analysis tool.
• Monitor mode scans all channels and reports to the WLC for security and intrusion detection — no external tool needed.

7. If WLC is deployed in WAN, which AP mode is best for local network and how?

Answer:

When the WLC is deployed in the WAN (remote/cloud location), the best AP mode for the local network is FlexConnect mode (previously known as Hybrid-Remote Edge AP or H-REAP).

Why FlexConnect is Best:

FlexConnect allows the AP to locally switch client traffic without sending it all the way to the WLC over the WAN. This is critical because:

• Without FlexConnect, all client data would have to travel through the CAPWAP data tunnel to the remote WLC and back — wasting WAN bandwidth and adding significant latency.
• With FlexConnect, the AP can forward client data directly to the local network (local switching), reducing WAN dependency.

How FlexConnect Works:

1. **Connected State (WLC reachable):**
   - Control traffic (authentication, configuration) goes through the CAPWAP control tunnel to the WLC over WAN.
   - Client data is switched locally at the AP — it does not go to the WLC.
   - The WLC still handles authentication (802.1X/RADIUS) and policy.
2. **Standalone State (WLC unreachable — WAN link down):**
   - The AP continues to serve clients using its locally cached configuration.
   - Clients can still connect and access local resources.
   - Authentication uses locally cached credentials if the RADIUS server is also remote.

This makes FlexConnect ideal for branch offices, remote sites, or any deployment where the WLC is far away over a WAN link.

8. What are the challenges if deploying autonomous APs (more than 50) in a large network like a university

Answer:

Deploying more than 50 autonomous APs in a large network like a university presents significant challenges:

1. **Management Complexity:**
   Each autonomous AP must be configured individually — there is no central controller. With 50+ APs, making a simple change (like updating the SSID password or security policy) requires logging into each AP separately, which is time-consuming and error-prone.
2. **Inconsistent Configuration:**
   Without centralized management, different APs may end up with slightly different configurations, leading to inconsistent user experience across the campus.
3. **No Seamless Roaming:**
   Autonomous APs do not support seamless Layer 2 roaming. When a student moves from one building to another, they may experience disconnection and re-authentication, disrupting ongoing sessions.
4. **No Centralized Security Policy:**
   Enforcing uniform security policies (WPA3, 802.1X authentication, ACLs) across all APs is difficult. Each must be configured manually and consistently.
5. **RF Management:**
   Autonomous APs cannot coordinate channel selection and transmit power automatically. This leads to channel interference and coverage gaps in high-density environments like lecture halls.
6. **No Rogue AP Detection:**
   Autonomous APs lack the ability to detect rogue APs or unauthorized devices on the network.
7. **Firmware Updates:**
   Updating firmware on 50+ APs individually is a major operational overhead and may lead to version inconsistencies.
8. **Scalability:**
   Adding new APs or expanding coverage requires significant manual effort compared to a controller-based deployment.

9. What happens on wireless client connected to Lightweight AP in local mode if WLC goes down

Answer:

When a Wireless LAN Controller (WLC) goes down and the Lightweight AP is in Local Mode, the following happens:

1. **CAPWAP Tunnel Drops:**
   The AP detects that the WLC is unreachable when keepalive (Echo Request/Response) messages stop being acknowledged after the retry threshold is exceeded.
2. **Existing Clients are Disconnected:**
   In Local mode, all client data is forwarded through the CAPWAP data tunnel to the WLC. When the WLC goes down, the data tunnel collapses. All currently connected clients lose their wireless connection immediately.
3. **AP Attempts Failover:**

The AP attempts to connect to a secondary WLC (if configured). If a secondary or tertiary WLC is available and reachable, the AP will re-establish the CAPWAP tunnel with it and clients can reconnect.

4. **No Local Switching:**
   Unlike FlexConnect mode, Local mode APs cannot switch traffic locally. Without the WLC, the AP cannot forward any client data — there is no fallback local operation.
5. **AP Enters Discovery State:**
   If no backup WLC is available, the AP will continuously try to discover and join a WLC. During this time, the AP radio may stop beaconing or clients attempting to connect will be rejected.
6. **SSIDs May Go Down:**
   The SSIDs broadcast by the AP will typically go offline since the AP cannot operate independently in Local mode.

Summary: In Local mode, WLC failure = complete loss of wireless service on that AP. This is why FlexConnect mode or having a redundant WLC (HA pair) is recommended for critical deployments.